



Guidelines

Tracheostomy Care

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Nursing

Introduction

This guideline outlines the procedures and responsibilities of individuals and the health service organisation in relation to nursing management and care of patients with a tracheostomy. It ensures the delivery of a uniform, integrated multidisciplinary approach with management that is consistent with best practice.

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☒	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGH Mission & Values	☒	Staff Safety	☒
Other:	☐		

Table of Contents

Introduction	1
Risk Statement.....	1
1.0 Principles of Tracheostomy Care	2
2.0 Suctioning	9
3.0 Monitoring for Occlusion of Tube	11
4.0 Humidification	12
5.0 Dressing / Stoma Care	13
6.0 Percutaneous Tracheostomies	14
7.0 Mini Tracheostomy.....	15
8.0 Adjustable flange Tracheostomy Tube.....	15
9.0 Changing Tracheostomy Tube.....	18
10.0 Tracheostomy Tubes with Removable Inner Cannula	20
11.0 Tracheostomy De-Cuffing.....	21
12.0 Removal of Tracheostomy (Decannulation).....	24
13.0 Intra / Inter-Hospital Transport of Tracheostomy Patients	26
References	27
Appendix 1: Tracheostomy Care, Cuff Deflation and Removal Regimen	30

1.0 Principles of Tracheostomy Care

1.1 Safety Information

The following equipment must be readily available at all times:

- Resuscitator bag and tubing
- Catheter mount (Bodai connector)
- Artery forceps
- Tracheostomy wire[‡] (available in resuscitation trolley)
- Appropriate resuscitation equipment
- Capnography available in event of emergency¹⁻⁴:
 - Waveform capnography used in ICU (from ICU transport trolleys)
 - Portable waveform Capnography device available from anaesthetics / ICU MET Bags

The following equipment must be available at the bedspace, and during procedures/transfer at all times for a patient with a tracheostomy:

- Spare tracheostomy tubes one of the same size and one smaller⁵⁻⁹
- Operational suction (checked at least daily) - may be portable or wall mounted with appropriately sized catheter attached and Yankauer sucker available.^{2-4,6,7}
- Tracheal dilators⁵⁻⁹
- Tracheostomy mask^{6,7}
- Spare Tracheostomy dressing and Velcro tapes
- Oxygen⁶
- Cuff manometer and 10mL syringe^{6,7}
- Stitch cutter^{1,2} and scissors^{2,6}
- Lubricating jelly²
- Sterile water³ or 0.9% saline¹ ampoules and clean kidney dish for inner cannula^{2,3}
- PPE [non-powdered latex free gloves, apron, facial protection]^{5,7-9}

Tracheostomy or Laryngectomy Emergency Algorithms and Bed Head signs are to be displayed at all times until discharged (refer to pages 5-8).

Tracheostomy Management, Resources and Contact information for specialist tracheostomy advice can be found on page 9.

Prior to assuming responsibility for the management of a patient with a tracheostomy the nurse and treating team must receive clinical handover inclusive of:

- Tube insertion date, reason for insertion, type and method of insertion and any complications during insertion, size and method secured (tapes and/or sutures).
- Relevant clinical history including baseline vital signs and clinical findings.
- Oxygenation status and supplemental oxygen requirements.
- Secretions (amount, colour, odour and consistency)³, humidification and suction requirements.

The following must be assessed at commencement of shift³ and documented at least once per shift in the inpatient notes / Nursing Care Plan (MR 855):

- Patient comfort and respiratory status/airway patency, cough effort and ability to swallow
- Check tapes are secure
- Minimal occlusive cuff pressure (if cuff present) (usually 18-25cmH₂O.)
- Assess tracheostomy dressing for ooze, bleeding, secretions
- Sputum characteristics (Colour, Volume, Consistency, Odour)
- Inner cannula patent (see section 9: Tracheostomy tubes with removable inner cannula).
- Bedside equipment additional to the above is present and in working order (i.e. oxygen).

Observe the five moments for hand hygiene and aseptic non-touch technique when manipulating tracheostomy tubes. Follow the risk assessment outlined in the SCGOPHCG Infection Prevention and Control Manual - Aseptic Technique Policy. This must include use of appropriate Personal Protective Equipment (PPE) to prevent contamination and mucosal or conjunctival splash injuries.^{11,21,23}

1.2 Nursing Management

- It is recommended that all nursing staff caring for patients with a tracheostomy complete the e-learning available at SCGOPHCG Learning Hub/ 'Tracheostomy Safety e-learning'.
- Only a nurse experienced in tracheostomy tube management may escort a patient with a tracheostomy when not accompanied by a medical officer. †
- The cuff provides a seal maintaining positive pressure ventilation and reducing aspiration of secretions. The cuff pressure should be checked every 8 hours^{3,9-11} and maintained below 25cmH₂O.^{1,3,9} Impairment of tracheal capillary blood flow will be caused by pressures above this. If air leak occurs at or above 25cmH₂O inform a senior medical officer as a larger tube may be required.^{1,3}
- Patients with tracheostomies may experience difficulties swallowing as a result of the cuff occluding the oesophagus.^{1,3} Referral to speech pathology should be considered for all patients not expected to rely solely upon enteric tube feeding.^{1,3,11}
- Patients undergoing tube feeding must be positioned with the head of the bed elevated to 45 degrees¹⁰ and ideally all patients with artificial airways should be nursed in the semi recumbent position (30-45 degrees) at all times.¹²
- Patients with tracheostomies should be encouraged to exercise and mobilise early to facilitate decannulation and improve lung function.¹²
- Oral care is important in reducing the risk of infections as a result of aspiration of oral secretions and development of biofilm on artificial airways by migrating oral flora. Teeth should be brushed at least twice daily³ and ensure adequate oral lubrication. The oral cavity should be moistened with a large cotton tip, or toothette swab every 2 hours, or as requested by the patient.† The oral cavity should be suctioned as required to remove pooled secretions above the tracheostomy cuff.^{10,11}
- Chlorhexidine mouth rinse or gel is recommended as being effective in reducing colonisation with nosocomial organisms and reducing the incidence of ventilator-associated pneumonia.^{3,11,13-15}

- **TRACHEOSTOMY EMERGENCY MANAGEMENT ALGORITHM** (refer to Diagram 1)
Must be displayed at all times at the head of bed (excluding ICU/theatre) until discharged. (In ICU, this is to be attached to the resuscitation and difficult airway trolleys.)† A copy must also be included in the inpatient notes.
 - **TRACHEOSTOMY BEDHEAD SIGN** (refer to Diagram 2)
Must be displayed at all times at the head of bed (excluding ICU/theatre) until discharged. A copy must also be included in the inpatient notes.
 - **LARYNGECTOMY EMERGENCY MANAGEMENT ALGORITHM** (refer to Diagram 3)
Must be displayed at all times at the head of bed until discharged. A copy must also be included in the inpatient notes.
 - **LARYNGECTOMY BEDHEAD SIGN** (refer to Diagram 4)
Must be displayed at all times at the head of bed until discharged. A copy must also be included in the inpatient notes.
- All signs are to be removed and cleaned appropriately post-discharge.

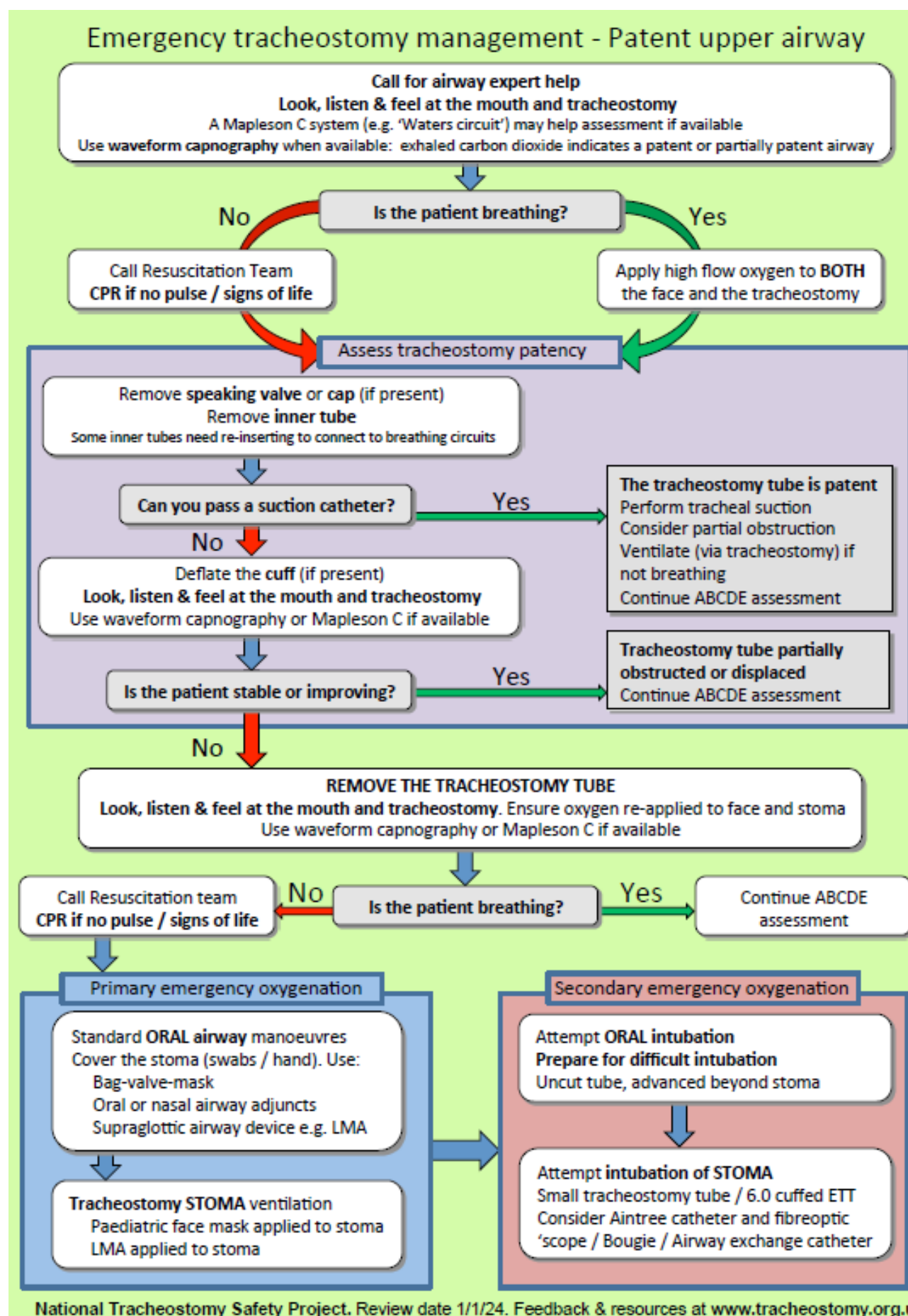
Diagram 1: Tracheostomy Emergency Management Algorithm¹⁶

Diagram 2: Tracheostomy Bedhead Sign¹⁷

- Must be displayed at all times at the head of bed (excluding ICU/theatre) until discharged.

This patient has a

TRACHEOSTOMY

There is a potentially patent upper airway (Intubation may be difficult)

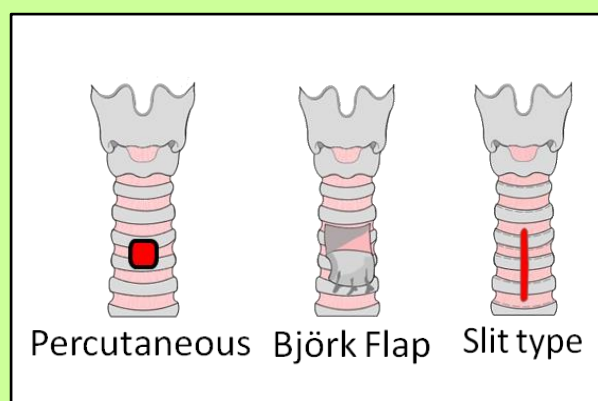
Surgical / Percutaneous

Performed on (date)

Tracheostomy tube size (if present)

UMRN

Notes: Indicate tracheostomy type by circling the relevant figure.
Indicate location and function of any sutures.
Laryngoscopy grade and notes on upper airway management.
Any problems with this tracheostomy.



Emergency Call: MET Dial '55' Duty Anaesthetist: Dect 71242 ICU Sr Reg: Dect: 76891 ENT: via switch '91'

www.tracheostomy.org.uk

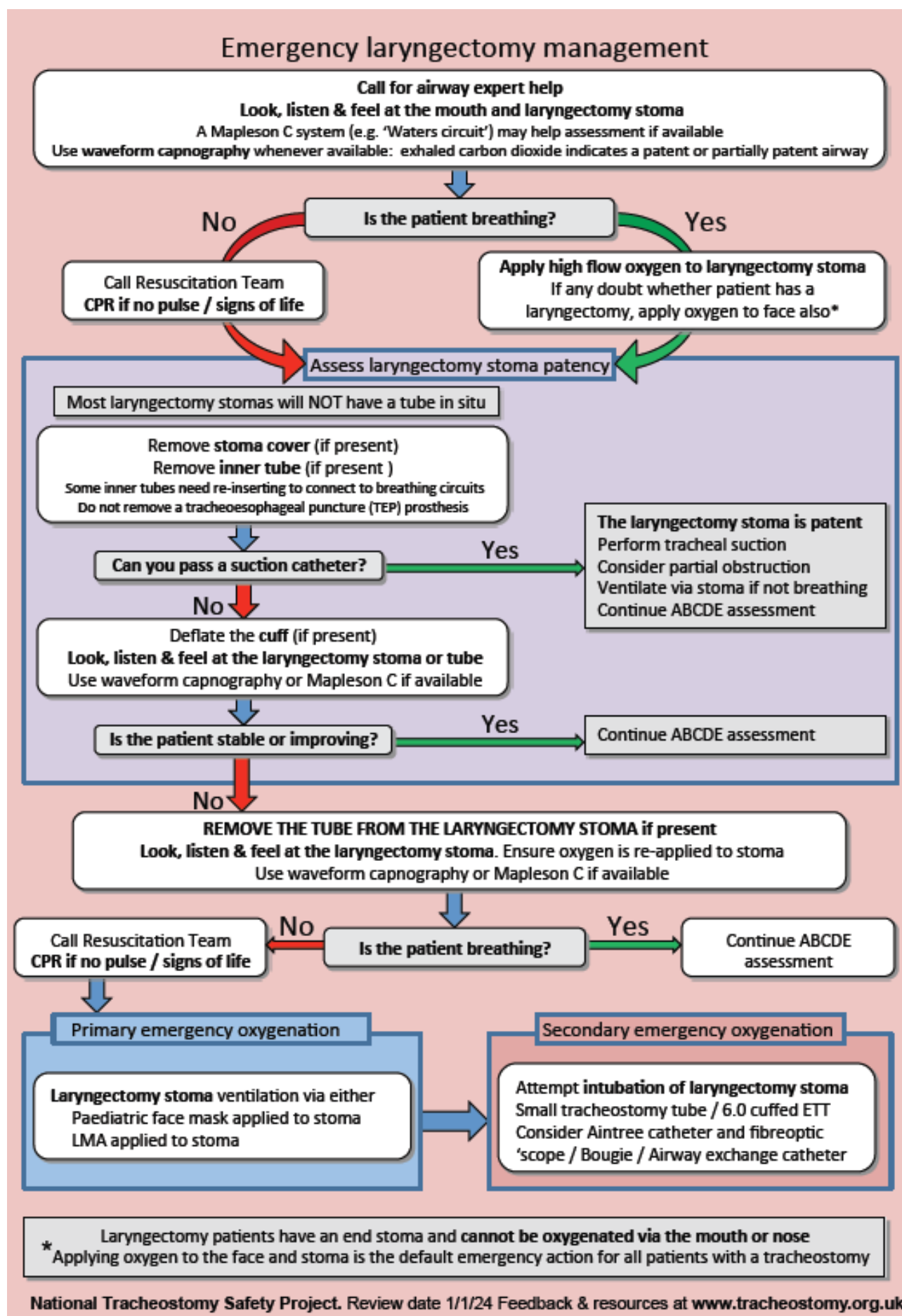
Diagram 3: Laryngectomy Emergency Management Algorithm¹⁸

Diagram 4: Laryngectomy Bedhead Sign¹⁹

Must be displayed at all times at the head of bed until discharged. (In ICU, this is also to be attached to the resuscitation and difficult airway trolleys.)[†]

This patient has a

LARYNGECTOMY

and CANNOT be intubated or oxygenated via the mouth

Follow the LARYNGECTOMY algorithm of breathing difficulties

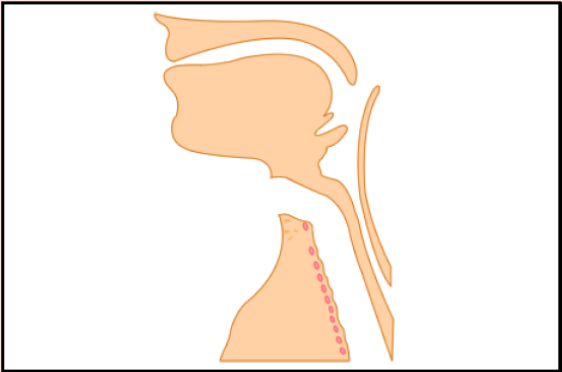
Performed on: (Date)

Tracheostomy tube size (if present)

UMRN

Notes:

There may not be a tube in the stoma.
The trachea ends at the neck stoma.



Emergency Call: MET DIAL '55' Duty Anaesthetist: Dect 71242 ICU Sr REG: Dect 76891 ENT: via switch 91

Adapted with permission from: www.tracheostomy.org.uk

TRACHEOSTOMY MANAGEMENT RESOURCES

(Ensure Medical Team, Ward Physio and Speech Pathology are informed of admission or transfer)

For details, refer to relevant Nursing Practice Guideline and Section.

GENERAL NURSING CARE	WEANING, SWALLOW, SPEECH AND DECANNUATION	RESPIRATORY CONCERNS	SUDDEN DETERIORATION	AFTER HOURS CONTACTS
• SEE RELEVANT NPG AND SECTION • For advice contact Shift Coordinator and/or SDN • For further advice contact Ward CNS. • For specialist nursing advice contact: G64 CNS Dect # 73707 ICU CNS Dect # 76888	• Liaise with CNS, Medical Team, Speech Pathology and Physiotherapy regarding weaning regimen. • No oral feeding without medical review. • Not for decannulation until cuff deflated for a minimum of 24 hours (see sections 10 & 11).	• INCLUDING BUT NOT LIMITED TO: ↑RR, work of breathing, shortness of breath etc. ↓SpO₂, mental state • CHECKS/ACTIONS MAY INCLUDE, BUT NOT LIMITED TO: Tube patency Tube position Inner cannula Adequate oxygen Adequate suction Patient position Humidification Administer nebuliser Activate MET call if warranted. • NOTIFY MEDICAL TEAM, WARD PHYSIO, AND SHIFT COORDINATOR • NOTIFY CNS IF ONGOING CONCERNS	• INCLUDING BUT NOT LIMITED TO: Blocked Tracheostomy Unplanned decannulation Acute Tracheomalacia • ACTIVATE MET CALL Emergency bell Dial 55 Gather resuscitation equipment Oxygen masks applied to both face and tracheostomy. Continue HBLS activities and care as per relevant NTSP Tracheostomy or Laryngectomy Emergency Algorithms.	• After Hours CNS Dect # 76231, 72997, 74478 • ICU Shift Coordinator Dect # 71180 • ICU Senior Registrar Dect # 76891 • Duty Anaesthetist Dect # 71242

2.0 Suctioning

2.1 Safety Information

Dangers of Suctioning:

- Hypoxia^{1,3,20,21}, bronchospasm, and alveolar collapse^{3,20,21}
- Trauma to mucous membrane^{1,20,21}
- Bleeding¹
- Tube Migration²⁰
- Cardiac arrhythmia and cardiovascular instability (bradycardia, hypertension)^{20,21}
- Increased intracranial pressure and reduction in cerebral blood flow (with coughing)²¹
- Decreased intracranial pressure (with prolonged continuous suction)
- Introduction of infection^{1,21}
- Patient Anxiety²⁰
- Suction catheters passed into the trachea can paralyse the cilia and increase the amount of secretions³
- Patient and staff become dependent on suction, instead of staff encouraging patient to cough secretions into the tube[†]

Ensure that tracheostomy tube is not inadvertently / partially withdrawn with removal of suction catheter[†]. Retention tapes around the neck should be securely fastened.⁹

Failure to advance the suction catheter may indicate that the tube is blocked, and appropriate senior staff must be informed immediately.^{1, 3}

2.2 Nursing Management

CALL FOR ASSISTANCE and follow tracheostomy emergency management algorithm if signs of respiratory obstruction are evident, for example:

- Noisy or absent air flow
- Dyspnoea
- Stoma or rib retraction, quickening pulse rate
- Inability to maintain patient's normal oxygen saturation
- Bradycardia
- Frequency of suctioning is determined by the clinical need of the patient.^{3,5,11,22,24} Closed suctioning devices should be utilised where available as the system of choice for patients with a tracheostomy who require suction.
- Suctioning should be performed:
 - At least 8/24 and PRN to remove retained secretions in tracheostomy tube and prevent occlusion in all patients with a tracheostomy, even in patients with a strong cough.²²
 - Whenever the patient is unable to clear own airway (e.g., patients with a weak cough); more frequent, deeper suction will be required.¹¹ This may be indicated by visible, palpable, or audible secretions^{11,20,21,25}, or 'saw-tooth' pattern on the flow-loop pattern on the monitor of the ventilator or CPAP machine.^{11,21}
 - Upon observation of desaturation, rising inspiratory pressures, decreased inspiratory volumes, increased inspiratory rate, and increased work of breathing or coarse breath sounds on auscultation.^{11,21}
 - Prior to cuff deflation²⁵, tube change²⁵ and decannulation²⁰.
 - Upon patient request.³
 - Suction apparatus and tubing must be checked for functionality each shift^{11, 13}, and, changed daily and PRN. (Refer to NPG No18 Suction, for care of suction equipment).

- Regulated suction (at a pressure of 80-150 mmHg) is recommended during suctioning of tracheostomy tube^{3, 20,21,24,26}. Regulated suction apparatus may be obtained from the Hospital Equipment Service or some ward areas may keep devices locally.
- Unregulated wall suction available in ward areas may be used if regulated suction cannot be sourced.
- **Closed suction** is routinely used in ICU for patients receiving Non-Invasive Ventilation (NIV) / Invasive Ventilation. Standard precautions are required when performing this type of suction.
- **Open suction technique** (when neither a Bodai or closed suction device is in use) is used in ward areas and is considered an aerosol generating procedure (AGP). PPE must include protective eyewear and N95/P2 respirator, non-sterile gloves and gown / apron when an open suction technique is used.^{11,21,26}
- Pre-oxygenation with 100% oxygen is recommended in patients who are at risk of hypoxia during suction.^{3,20-22,24}
- A 10 ml syringe can be used for emergency re-inflation/deflation of cuff. Cuff pressure must be checked with a tracheostomy manometer as soon as possible. † Cuff Pressure should never be more than 25cmH₂O. ^{1,3,9}
- The instillation of 0.9% saline to loosen secretions prior to suctioning is **not** recommended.^{21,27,28} Normal saline instillation may introduce bacteria to the lower airway tract.²²
- The suction catheter should be less than half the size of the inner diameter of the tube.^{11,21,22,24,26}
- Patient assessment before, during and post suction should include an evaluation of the effects on the patient's pre-suction signs and symptoms. This should include monitoring of cardiac rate and rhythm, blood pressure, pulse oximetry, airway reactivity, tidal volumes, peak airway pressures, and intracranial pressure.²¹
- Total suction time should not exceed 10-15 seconds.^{20,22}

2.3 Procedural Information

To set the Amvex device (Figure 1) to deliver regulated suction:

1. Attach to wall suction outlet
2. Connect yellow suction tubing between filter outlet and suction canister
3. Turn grey selection switch to 'REG'
4. Occlude suction tubing and manipulate grey dial until pressure indicator sitting between 80 – 150 mmHg. This should be completed at the commencement of each shift.
5. Move grey selection switch to 'OFF' once suctioning completed.
6. 'FULL' is unregulated suction.



Figure 1. Amvex Regulated Suction device

Requirements

- As per mandatory tracheostomy equipment (refer Section 1, Principles of tracheostomy management)
- Sterile suction catheters with "Y" connection (appropriate size)
- Container with water for rinsing suction tubing
- PPE (protective eyewear and N95/P2 respirator, non-sterile gloves and gown / apron)

Procedure**Open Suction technique**

1. Explain the procedure to the patient.
 - Patients should be given clear information regarding the suction procedure including: the need for suction and the consequences of not suctioning.
 - This should be repeated at each suction procedure as they may not recall previous instructions.²¹
1. Pre-oxygenate patient with 100% oxygen if required.^{3, 20-22,24}
2. Turn on and check suction.
3. Perform hand hygiene.
4. Open "Y" suction catheter packet using a non-touch technique, taking care not to touch the catheter.
5. Don non-sterile gloves immediately prior to procedure.²⁹
6. Attach sterile suction catheter to suction tubing and remove catheter from wrapping.
2. If the patient has loose secretions that can be coughed to the end of the tube, then "shallow suctioning" may be sufficient³:
 - Insert the suction catheter into the tip of the tracheostomy tube and clear secretions only from the device itself, by advancing to a depth of 3cm.
3. If the patient is unable to cough or if auscultation indicates secretions distal to the tip of the tube, then deeper bronchial suctioning may be required.
 - Gently insert catheter during inspiration to a depth of 10-15 cm with "Y" valve open until cough reflex is elicited and/or resistance is encountered at the carina.^{3,21,22,26}
 - Withdraw catheter 1- 2cm^{21, 22, 24, 26}. The catheter is then withdrawn smoothly without rotating³, whilst maintaining continuous suction. Suction should only be applied as the catheter is being withdrawn^{3, 21}. The entire process should not exceed 10-15 seconds.^{20,22}
4. Discard gloves and catheter.
5. Use catheter and glove once only.
6. Perform hand hygiene and don clean non-sterile gloves
7. Perform oral suctioning and clear supraglottal secretions using Yankauer sucker as necessary.
8. Rinse suction tubing with tap water.
9. Remove gloves and perform hand hygiene.
10. If patient receiving O₂ therapy, continue oxygen at set flow rate.
11. On completion reassess patient's chest movement, breath sounds and haemodynamic status.^{21,29}
12. Document procedure and findings (including volume, colour, consistency, and frequency) in patient's medical record.^{20,21}

3.0 Monitoring for Occlusion of Tube**3.1 Nursing Management**

- Observe patient's respiratory status note patient respiratory rate and effort.
- If only partial occlusion is present, suction and nebulisation may be adequate.
- If signs of tube occlusion are present:

- Noisy or absent air flow,
 - Dyspnoea,
 - Stoma or rib retraction, quickening pulse rate.
- **Call for assistance or activate MET call if patient's condition significantly compromised. Continue care as per HBLIS and the tracheostomy emergency management algorithms.**

3.2 Procedural Information

Procedure

In the event of a presumed occlusion:

Please refer to [Diagram 1: Tracheostomy Emergency Management Algorithm](#)

1. Remove inner cannula and any speaking valves if in-situ.
2. Pass a suction catheter through the tracheostomy into the trachea; if it passes, the tube is considered patent.¹⁶ Perform suction.¹⁶ Continue A – E assessment¹⁶ and HBLIS emergency procedures as per standard algorithms.
3. Consider saline nebulisation to loosen mucoid plug[†].
4. If the catheter cannot pass, the tube may be partially or completely occluded. Deliver oxygen via Hudson mask to the patient's face.
5. Deflate cuff¹⁶ to allow patient to breathe around tube (Jaw support may be required)[†].

4.0 Humidification

4.1 Medical Requirements

Prescription for normal saline nebuliser and the frequency required.

4.2 Safety Information

Inform medical staff immediately if difficulty in passing a suction catheter or signs of airway occlusion present ^{1,3}.

Adequate humidification is essential for all patients with tracheostomies to improve comfort and, reduce the risk of infection, airway damage, retained secretions and tube occlusion.^{1,3,9,30}

Tracheal humidification may be provided by heated humidifier ^{11,30}, passive humidifier ^{11,30} [e.g., heat and moisture exchange filter (HMEF)], or saline nebulisers¹¹ via tracheostomy mask. A newly created tracheostomy will require greater humidification than an established one. Ensure adequate hydration of the patient^{1, 3, 11}.

4.3 Nursing Management

- Ensure optimal humidification maintained by either humidifier or saline nebulisation.^{11,30}
- Monitor amount and consistency of secretions. Increase humidification or add / increase the frequency of saline nebulisation if secretions are thick or patient having difficulty in expectorating secretions.^{1,3}
- If using a passive humidifier (e.g., HMEF), additional regular 0.9% saline nebulisation will be required to ensure optimal humidification.
- Heated water bath humidification should be used for all patients with thick secretions.¹ The passive humidifier (e.g., HMEF) must be removed when administering a normal saline nebuliser.[†]
- Ensure patient is adequately hydrated to prevent drying of secretions.^{1,3,11}

4.4 Procedural Information

Refer to: [SCGOPHCG CPG Oxygen Therapy](#)

- Humidification for set up and use.
- Nebulisation for additional requirements and safety information.

5.0 Dressing / Stoma Care

5.1 Safety Information

Only pre-cut keyhole dressings are to be used to reduce risk of foreign body aspiration or lodgement within the stoma^{1, 3, 11}.

Over-inflation of cuff may result in localised ischemia, tracheal stenosis, fistula formation, erosion or tracheomalacia (collapse of the tracheal wall)³.

Strict management of the stoma is essential as maceration will occur if wet/soggy dressings remain in contact with the skin resulting in wound degradation.³ The stoma should be kept clean and dry to prevent post-incision wound infection.^{1,3} Red, excoriated, or exudative wounds should be reviewed for potential infection and wound swabs collected and sent if necessary.^{1,3}

5.2 Nursing Management

- Two nurses are required to change tapes^{3,11,31}, and dressings,^{3,32,33} one of whom must be a Registered Nurse.^{31,32}
- Dressing to be changed daily and PRN.³⁴ Patients with copious secretions will require dressing changes more frequently.^{3,33,34}
- Permanent tracheostomies require inspection of the stoma at least daily for signs of infection and irritation.¹
- If cuffed tracheostomy tube is used, assess that cuff is inflated to the minimum occlusive pressure ($<25 \text{ cmH}_2\text{O}$ ^{1, 3, 9}) and check cuff pressure 8/24^{3, 9,11} and PRN, using a tracheostomy cuff manometer¹. Document in inpatient notes the minimum occlusive pressure.^{1,3}
- Where the skin has become excoriated, a thin hydrocolloid dressing may prevent further deterioration.³⁵

5.3 Procedural Information

Requirements

- As per mandatory tracheostomy equipment (Refer Section 1, Principles of tracheostomy management)
- Dressing pack
- Velcro tracheostomy tapes
- Pre-packaged polyurethane keyhole dressing
- Sterile sodium chloride 0.9% for irrigation^{32,36}
- Skin barrier film wipes^{32,36} (i.e. Conveen[®]) (optional)
- Wet Strength bag
- Sterile cotton tips
- PPE

Procedure

1. Explain the procedure to the patient.
2. Perform hand hygiene.
3. Clean trolley with combined detergent disinfectant wipe. Allow to dry.
4. Gather equipment.
5. Perform hand hygiene.
6. Open dressing pack on top of trolley and additional items.
7. Perform hand hygiene and apply PPE.³⁶
8. Loosen tracheostomy mask straps.
9. Remove soiled dressing while wearing clean non-sterile gloves.

10. Inspect stoma for signs of infection such as erythema, pain, exudate or dried secretions.^{32,36}
11. Remove gloves, perform hand hygiene, and don clean non-sterile gloves.
12. Place sterile towel in position on patient's chest.³⁷
13. Lift flanges with forceps and swab using cotton tip or gauze swabs.³⁶ Clean around stoma with moistened sterile cotton tips / gauze using aseptic non-touch technique, in a semi-circular motion from the centre to outwards.³⁵
14. Ensure gauze swabs are wrung out well to avoid fluid entering trachea[†].
15. Dry around stoma in same manner³⁶, in a semi-circular motion from centre to outwards.³⁵ Examine stoma for signs of erosion.³⁵
16. Change tapes PRN. Second nurse to support flanges with gloved fingers to prevent tracheostomy tube from dislodgement during tape removal.
17. Where possible ensure tube, flanges are well seated against skin. Secure clean tapes or Velcro through flange ends. Tie with knots on both sides of patient's neck if cotton tapes are used.³¹ The flanges of the tracheostomy tube are not sterile; therefore, new tapes may be applied without the aid of forceps[†].
18. Avoid over-tightening the securement device. It should be possible to insert 1-2 fingers under the tape.^{3,11,31}
19. Inspect skin under tapes / flange to ensure good skin integrity is maintained.
20. If prone to secretions and moisture, consider use of barrier film wipe around stoma. Do not allow wipe to touch stoma. Allow to dry before applying new dressing.
21. Position new keyhole dressing around tracheostomy tube using forceps. Other specialised dressings, such as foam dressing may be used as indicated.
22. Check cuff pressure once tube well seated and secured.
23. Wipe and replace tracheostomy mask and continue oxygen therapy as prescribed.
24. Dispose of waste appropriately and remove gloves.
25. Perform hand hygiene.
26. Document all dressing and tape changes in the integrated progress notes.^{3,31,32,36}

6.0 Percutaneous Tracheostomies

6.1 Medical Requirements

A percutaneous tracheostomy is one created percutaneously (through the skin) via wire guided dilation into the trachea. There is no surgical incision of the trachea, so the initial tube change may be difficult as the stoma will be tight.³

In the event of percutaneous tube dislodgement or removal, the retracted tissues recoil making replacement of the tube more difficult. The initial tube change requires an introducer to guide tube placement and should only be performed by a trained proceduralist.^{1,3}

The inserting Medical Team performs the first change of a percutaneous tracheostomy tube 10-14 days post insertion ‡.

6.2 Safety Information

All appropriate equipment should be readily available. This must include equipment for emergency management of the airway and reinsertion of the tube.⁴

A gum elastic bougie or oxygenating intubation catheter is to be available for the insertion procedure ‡.

A spare tracheostomy guide wire must be always readily available in the event of emergency for all percutaneous tracheostomies. ‡

For 24 hours following initial tube insertion, a bronchoscope must be readily available[‡] (It may be obtained via the ICU or ENT registrars).

For a percutaneous tracheostomy change, monitoring must include pulse rate, blood pressure, pulse oximetry, ECG and capnography for confirming ventilation. The equipment must be checked for completeness and functionality.⁴

6.3 Nursing Management

Routine care of percutaneous inserted tracheostomy tubes is identical to that of other tracheostomies.

7.0 Mini Tracheostomy

7.1 Medical Requirements

A Medical Officer must perform insertion.

7.2 Safety Information

Mini tracheostomies are not appropriate for providing artificial ventilation except during an emergency.[†]

A Mini tracheostomy does not provide an airway and is contraindicated in patients with hypoxia or hypercapnoea. Therefore, patients unable to self-ventilate effectively are not appropriate candidates for mini tracheostomy, except in some emergency situations[†].

7.3 Nursing Management

- Mini - tracheostomies are a small-bore cannula (4.0mm internal diameter) generally inserted into trachea of patients who have secretion retention issues to allow for regular suctioning.^{3,37,38}
- Oxygen can be provided via a mini tracheostomy in emergency situations³. The cannula is also suitable for oxygen therapy³, and nebulised drugs.
- Patients should be able to breathe normally through their mouth / nose during suctioning procedures in the presence of a patent airway.[†]

7.4 Procedural Information

Requirements

- As per mandatory tracheostomy equipment (refer Section 1, Principles of tracheostomy management)
- Sterile suction catheters with "Y" connection (appropriate size)
- Container with water for rinsing suction tubing
- PPE

Procedure

1. Suctioning is performed in the same manner as outlined for a standard tracheostomy.[†]
2. A size 10 FG or smaller suction catheter must be used.³⁷
3. Dressings (if required) are performed in the same manner as outlined for standard tracheostomy.[†]
4. Removal is performed in the same manner as outlined for standard tracheostomy (Disregard cuff deflation requirement).[†]
5. The mini tracheostomy must be capped or spigotted after suctioning.

8.0 Adjustable flange Tracheostomy Tube

8.1 Medical Requirements

A Medical officer must perform insertion

8.2 Safety Information

External length measurements to ensure correct placement is maintained, as a minimum once each shift

8.3 Nursing Management

- Each shift check and measure external length of tracheostomy (figure 2). Document on Tracheostomy regime:MR955.2 or ICU flow chart
- Each shift ensure the manufacture midline marking is central and in line with locking device (figure2), and if not midline inform medical officer.
- Inner cannula to be cleaned using specialised foam cleaning swabs ⁴⁴

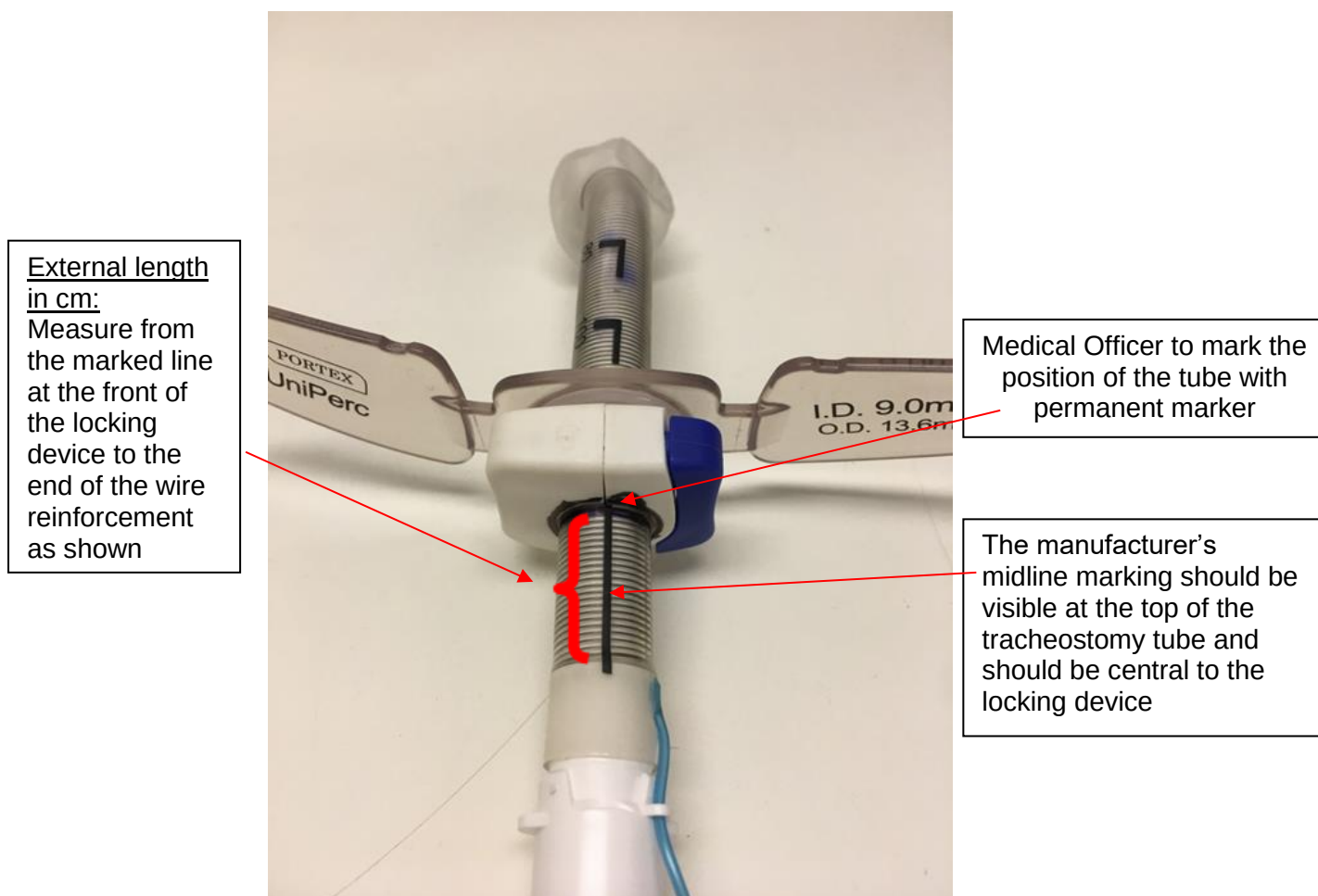


Figure 2

8.4 Procedural Information

Requirements

- As per mandatory Tracheostomy Equipment (refer Section 1, Principals of Tracheostomy management)
- Sterile suction catheters with “y” connection (appropriate size) or closed suction system where appropriate in ICU.
- Container with water for rinsing suction tubing
- PPE

Procedure:

1. Suctioning is performed in the same manner as outlined for standard Tracheostomy
2. A size 10 FG or larger suction catheter must be used
3. Dressings (if required) are performed in the same manner as out lined for standard tracheostomy.
4. Removal is performed in the same manner as outlined for standard tracheostomy

9.0 Subglottic Port (Portex SuctionAid) Tracheostomy Tubes

The Suctionaid tracheostomy tube allows for removal of above cuff subglottic secretions and may be used for above cuff vocalisation.

Safety Information

- Do not use the Suctionaid® tracheostomy tube for voicing until 72 hours post insertion to avoid subcutaneous emphysema.
- If required, intermittent low pressure wall suction can be used when clearing above cuff secretions via the suctionaid line. Do not apply continuous suction as this may cause trauma to the tracheal wall.

Nursing Management

- Retain thumb control valve in clean packaging for use if wall suction required for subglottic secretion removal.
- All patients with subglottic Suctionaid tracheostomy tube should have routine removal of above cuff secretions to reduce the risk of the accumulation of bacteria laden secretions.
- The minimum recommended frequency of secretion removal is 8/24.

Procedural information

Voicing/Talking

The Suctionaid® tracheostomy tube facilitates voicing in the presence of an inflated cuff. Medical air (preferably) or oxygen is entrained through the suctionaid line. When the thumb control port is occluded, air is directed over the vocal cords.

The patient's ability to vocalise via the Suctionaid® tracheostomy tube is initially assessed by a Speech Pathologist who will determine if the patient is able to vocalise using this method.

Voicing via the Suctionaid® line can then be performed by others who have received instruction in use of this tube.

Do not use Luer Lock syringes for this purpose as they may crack the suction aid port.

Removing above cuff secretions Via syringe: (Preferred method)

- Insert 10 - 20 mL luer slip syringe into Suctionaid® Subglottic suction port.
- Gently pull the plunger towards you to collect the secretions.
- Remove syringe from port.
- Document amount and appearance of secretions removed from above the cuff

Removing above cuff secretions via regulated wall suction unit (if syringe unsuccessful):

- If required, intermittent low pressure wall suction can be used. Do not apply continuous suction above the cuff as this may cause trauma to the tracheal wall.
- Insert the thumb control valve into the Suctionaid® subglottic suction port.
- Connect the suction tubing to the thumb control valve.
- Set the wall pressure gauge to low pressure (150mmHg).
- Perform intermittent suctioning by briefly occluding the thumb control valve on the suction line. Remove thumb from the port and if further suctioning is required repeat procedure.
- Cease suctioning once the majority of secretions have been cleared from Suctionaid® subglottic suction port. Do not continue to suction for only minimal amounts of secretions due to the risk of trauma to the upper airway.

Blocked subglottic suction port:

Troubleshooting must NOT be used within 72hrs of a freshly formed tracheostomy stoma due to risk of surgical emphysema.

- If the subglottic suction line becomes blocked, insert 3-4ml air through the line via 10ml syringe.

- Alternatively inject 1ml sterile water into the subglottic suction line and then remove using the same syringe and discard.

10.0 Changing Tracheostomy Tube

10.1 Medical Requirements

Percutaneous Tracheostomies are to be reviewed by the inserting medical team at first change. Surgical Tracheostomies to be reviewed by team involved with insertion.

At time of review medical staff will decide if tube requires changing.

10.2 Safety Information

First Routine Tube Change (For All Non-Percutaneously Inserted Tubes):

- First routine tube change is to be performed by medical and nursing staff proficient in tracheostomy change^{1,10,25}.
- The tube should be changed within 5 to 7 days following surgical tracheostomy.^{1,11,25}
- A Medical Officer proficient in intubation must be present on the ward and available to assist if required.¹¹

First Routine Tube Change (For Percutaneously Inserted Tracheostomy):

- The inserting medical team must perform the first tube change. Following percutaneous tube removal, the retracted tissues recoil making replacement of the tube more difficult than surgical tracheostomy tube change^{2,3}.
- Any initial tube change must be approved by an ENT or ICU consultant.
- If possible, the tube should not be changed for 10 to 14 days following a percutaneous procedure. ‡ Premature change of tube increases the risk of false track formation or incorrect placement.³⁹

Subsequent Tube Changes:

- For subsequent changes, Medical Officer must be informed and present on the ward. †
- The procedure may be performed by two nurses, one of whom must be a registered nurse experienced in the procedure. ‡
- Request urgent medical review if any suspicion of malposition of tube, creation of a false passage, surgical emphysema, breathing difficulties or hypoxaemia

10.3 Nursing Management

- **Tracheostomy tubes without removable inner cannula** need to be reviewed at 7 days and earlier if signs of airway obstruction present.³
- **Tracheostomy tubes with inner cannula** can be changed every 30 days or earlier if any signs of airway obstruction present.^{1,3}
- Sutures should be removed on day 5 post insertion unless there is a documented medical reason for them to remain insitu.³
- Tube placement must be confirmed with capnography following insertion.¹⁻⁴
- Patient must fast 4 hours prior to this procedure.^{1,11}
- Some slight bleeding is normal following a tracheostomy tube change. If bleeding persists notify Medical Officer. †
- Record date, time of change, ease of insertion and any problems encountered in medical record.^{1,11}

10.4 Procedural Information

Requirements

- As per mandatory tracheostomy equipment (refer Section 1, Principles of tracheostomy management.)

- Pulse oximetry
- Dressing pack
- Lubricant
- Sterile sodium chloride 0.9%
- PPE

Note: additional emergency equipment that must be present during a planned tracheostomy change or decannulation includes:

- Oxygen and selection of masks including non-rebreather, Hudson and tracheostomy masks.
- Resuscitation bag and appropriate sized mask
- Intubation equipment (including Bougie or oxygenating intubation catheter) ‡ and a range of tracheal tubes.
- Laryngeal mask
- Tracheostomy cuff manometer

Procedure

1. Explain the procedure to the patient.
2. Perform hand hygiene.
3. Position patient supine, remove / adjust pillow to allow the neck to be a neutral position (i.e., not flexed or excessively extended).
4. Perform hand hygiene.
5. Aspirate nasogastric tube if in situ.^{1,3,11}
6. Pre-oxygenate the patient.
7. Suction oropharynx and tracheostomy.
8. **Perform hand hygiene and apply PPE.**
9. **Operator:** Open dressing pack and prepare equipment. Cuffed tube - check that cuff is patent on inflation, is symmetrical, has no evident leaks and the introducer is easily removed. Maximally deflate cuff and apply lubricant. Wring out gauze swabs.
10. **Assistant:** Remove / loosen tracheostomy oxygen supply. Remove soiled dressing while wearing clean non-sterile gloves.
11. **Operator:** Clean and dry stoma using swabs.
12. **Operator:** Perform hand hygiene and don nonsterile gloves.
13. **Assistant:** Cuts / removes tapes, deflates cuff, and removes tube. Perform hand hygiene.
14. **Operator:** Inserts new tube, removes introducer. (**Do not** rotate tube for insertion). Some resistance to placement of the new tube maybe experienced.
15. **Operator:** Supports tube insitu with fingers on flanges until secured.
16. **Assistant:** Slowly inflate cuff.
17. **Assistant:** Check position using StatCap[®] device or flow capnography.
18. Tie Velcro tapes.
19. Ensure airway is patent.
20. Remove gloves. Perform hand hygiene, don new gloves.
21. Repeat tracheal and oral suction.
22. Ensure oxygen flow rate is administered as prescribed.
23. Remove gloves and perform hand hygiene. Don non-sterile gloves.
24. Redress stoma.
25. Remove gloves and perform hand hygiene.
26. Check cuff pressure using the tracheostomy cuff manometer.
27. Dispose of waste and remove gloves. Perform hand hygiene.
28. Document procedure in patient medical record.

11.0 Tracheostomy Tubes with Removable Inner Cannula

11.1 Safety Information

Only use sterile sodium chloride 0.9% or sterile water as per manufacturer guidelines^{11,40,41}.

11.2 Nursing Management

- Some tracheostomy tubes have a removable inner cannula, which enables cleaning and removal in the event of airway obstruction if the tube becomes blocked.^{1,3}
- Disposable tracheostomy brushes should be used to clean inner tubes. These are to be disposed after each use.



- If spare inner cannula supplied, store in a clean kidney dish.^{1,3} Wrap in sterile gauze and cover with plastic bag to prevent dust collection.[†]
- A replacement inner cannula should be immediately available at the bedside in case of urgent removal due to obstruction.^{1,3,17,35}

Care of the Inner Cannula

- The inner cannula of a tracheostomy tube should be removed and cleaned or changed:
 - at least every four hours if the patient has copious secretions and during the first two days following insertion, or,
 - at least every 8 hours as routine.^{5,8,35}
- Cleanse with sterile sodium chloride 0.9% or sterile water as per manufacturer guidelines^{11,40,41}.
- If the tube is cuffed perform oropharyngeal suction prior to removal of the inner cannula.[†] (Refer section 1 Suctioning)

11.3 Procedural Information

Requirements

- As per mandatory tracheostomy equipment (refer Section 1, Principles of tracheostomy management.)
- Clean, dry, tracheostomy brush (single use item)
- Sterile gauze
- Clean kidney dish

Procedure

1. Explain the procedure to the patient.
2. Assess patient condition; ensure they are well oxygenated and able to tolerate procedure.⁴¹
3. Perform hand hygiene.¹¹
4. Don personal protective equipment and non-sterile gloves.^{11,41}
5. Stabilise tracheostomy flange with one hand.⁴¹
6. With other hand - grasp inner cannula and remove as per manufacturer's instructions.⁴¹
7. Remove inner cannula place in clean kidney dish.⁴¹ Clean using sterile sodium chloride 0.9% or sterile water as per manufacturer guidelines^{11,40,41} and tracheostomy brush¹¹. Do not leave the inner cannula to soak.³
8. At end of procedure rinse tube in sterile sodium chloride 0.9% solution or sterile water as per manufacturer guidelines.
9. Dry with sterile gauze or as per manufacturer guidelines¹¹.
10. Stabilise tracheostomy flange and reinsert cannula.
11. Lock cannula into position by turning 90° clockwise and click into place.
12. Remove PPE. Perform hand hygiene

13. Document on ICU Flow Chart or Tracheostomy Care Cuff Deflation and Removal Regime (MR 855.2).^{3,41}

Note: If spare inner cannula available, insert into patients' tracheostomy immediately after removal of dirty cannula. Clean dirty cannula and store in sterile gauze packet, within a clean kidney dish. †

12.0 Tracheostomy De-Cuffing

12.1 Safety Information

The patient must be medically stable¹¹

Two nurses should be present for the initial decuffing attempt.³

Consider ceasing enteral feeds prior to commencement of initial de-cuffing.

12.2 Nursing Management

Requirements to consider prior to commencing cuff deflation regime:

- Chest clear
- Afebrile^{1,3}
- Cough present^{1,3}
- Swallow/gag present, ability to clear secretions^{1,3}
- Minimal suction required
- Characteristics of secretions
- Oxygen requirements, respiratory requirements^{1,3}
- Sitting tolerance, fatigability, positioning limitations

Indications for cuff reinflation include:

- ↓ SpO₂ below 90%
- Increased coughing
- ↑ secretions / copious suctioning required
- ↑ respiratory rate
- Signs of respiratory distress – increased effort on inspiration and expiration, stridor
- High level of patient discomfort / tiring / restlessness/diaphoresis
- Evidence of discolouration in secretions after eating or drinking
- Change in medical condition, ↓GCS

If the goal is decannulation or weaning to an uncuffed tube increase decuffing time as tolerated by the patient¹¹. The patient should be able to tolerate the cuff deflated for a full 24-hour period (minimum) before removal can be considered.³

12.3 Procedural Information

Requirements

- As per mandatory tracheostomy equipment (refer Section 1, Principles of tracheostomy management.)
- Pulse oximetry
- Tissues
- Tracheostomy cuff deflation chart
- PPE
- 10ml syringe

Procedure[†]

1. Perform hand hygiene
2. Explain the procedure to the patient. Inform them they may experience increased coughing.
3. Document baseline observations. These include:
 - Vital signs - pulse rate, respiratory rate, SpO₂
 - Patient comfort and use of accessory muscles of respiration

- Signs of respiratory distress
 - Amount of suctioning required, type of secretions.
 - Level of consciousness
 - Time at start of cuff deflation
 - Time of cuff inflation, reason for cuff inflation
 - Total time duration of the cuff being deflated
 - Document above on ICU Flow Chart or Tracheostomy Care Cuff Deflation and Removal Regime (MR 855.2). (See Appendix 1)
4. Perform hand hygiene. Position the patient in an upright (>45°) position to encourage good lung function.³
 5. Perform hand hygiene.
 6. Don personal protective equipment.
 7. Suction prior to deflating the cuff³ or ask the patient to cough and clear tube. This removes any excess secretions that are sitting above the cuff. (Refer Section 2 – Suctioning)
 8. Use a 10ml syringe to fully deflate the cuff (Do not use the manometer to deflate the cuff as this method does not always remove all of the air from the cuff).
 9. Suction patient as required, performing hand hygiene following any further suctioning.
 10. Observe and document patient response during and following de-cuffing.
 11. Re-inflate cuff when planned time for deflation has been reached or if patient is unable to tolerate.)
 12. Use a 10ml syringe to fully inflate cuff and check it has been inflated to the desired pressure (<25cmH₂O^{1,3,9}) with a tracheostomy cuff manometer.
 13. Perform hand hygiene.

13.0 Speaking Valves

Medical Requirements

- The medical team must approve the use of a speaking valve/Cap.

Safety Information

- Do not use on patients who are mechanically ventilated.
- Do not use with a foam filled Tracheostomy as these can spontaneously reinflate.
- The Valve/Cap must never be attached to a cuffed tube when the cuff is inflated due to risk of death.
- Patient must be able to remove valve or cap independently.
- Cautious consideration is required when using one-way valves or caps for whom cognition, alertness and manual dexterity is limited as this may preclude them from being able to physically remove the one-way valve or cap in the event of respiratory distress.
- Valve/cap to be removed immediately if there are signs of respiratory compromise
- Remove Device prior to performing humidification or nebulisation.

Nursing Management

- Patient is to be assessed by speech pathologist or ENT ward staff for potential appropriate use.
- Patient should be closely monitored by a trained member of staff for the initial application of valve or Cap
- Remove Valve/cap if patient has difficulty Breathing or has any sign of stridor
- When Valve/Cap not in use store in a labelled yellow top container at bedside.

Procedural Information

1. Perform Hand Hygiene
2. Position the patient appropriately
3. Deflate the tracheostomy cuff is deflated using a 10ml syringe

4. Check oximetry of patient
5. Apply Valve/cap
6. Encourage patient to speak

One way Valve

- One way valve allows air to be inspired via the tracheostomy tube. The valve closes on expiration, thus directing the air past the vocal cords and out through the nose and mouth.
- They do not provide humidity
- Patient assessment is required to determine if humidification needs are met during one way valve use.

Cap

- The cap is placed at the end of the tracheostomy tube for the purpose of assessing upper airway patency and support the decision to proceed or forgo with decannulation.
- There is no airflow through tracheostomy tube when the cap is insitu. The patient will be reliant on their own respiratory system as airflow bypasses the tracheostomy tube.

14.0 Oral Feeding of Tracheostomised Patients

Medical Requirements

Medical Officers or nursing staff must request for patients to have a Speech Pathology swallow assessment. If appropriate, the patient can be fed orally according to Speech Pathology recommendations after assessment.

Safety Information

- Swallow safety can vary as a function of tracheostomy cuff status and presence of speaking valve. The tracheostomy cuff and/or speaking valve status must replicate the conditions under which Speech Pathology assessment occurred. In most cases, Speech Pathology will recommend oral intake while cuff deflated and speaking valve insitu.
- If concerns are raised at any time by medical, physiotherapy or nursing staff regarding the patient's ability to cope with ongoing feeding, all oral feeds will be withheld until Speech Pathology have reviewed the patient

Nursing Management

- During assessments by the Speech Pathologist, the tracheostomy cuff may require deflation for several short periods. Perform oropharyngeal and tracheal suction prior to cuff deflation.
- An observational fatigue assessment must be conducted prior to each oral feed. If the patient is drowsy, tired, tachypneic or uncooperative, delay feeding until the problem is rectified (There is a high risk of aspiration with these conditions).
- Position the patient in a high sitting position in bed or sitting upright in a chair.
- Minimise risk of aspiration by:
 - Sitting patient upright during feeds and for 30 minutes post oral intake.
 - Being aware of patient's fatigue level during oral intake.
- Assess for nausea and treat proactively. Delaying oral intake whilst treatment initiated may be indicated.
- Cuff deflation is not appropriate if high risk of vomiting is present. If patient vomits during cuff deflation, cease any oral intake, and re-inflate cuff.
- Do not exceed or change oral intake restrictions unless recommended by the Speech Pathologist.
- Nutrition via enteral route or TPN may be continued to ensure complete nutritional requirements are met. Oral intake must be accurately recorded to assist the Dietitian in their assessment of nutritional needs.
- Physiotherapists and Speech Pathologists may perform cuff deflation - ideally the initial trial will be done jointly.

- Staff experienced in tracheostomy management may provide ice chips to patients with tracheostomy in moderation to the patient for comfort.
- If foods other than ice-chips are desired, a formal Speech Pathology assessment must be completed, and appropriate foods documented prior to administration.
- Only those foods and fluids approved by the Speech Pathologist are to be given to the patient.
- If cuff deflation is part of the feeding and / or communication regime, document patient's response and outcome in the inpatient notes.

Procedural Information

There is a high prevalence of silent aspiration associated with tracheostomy. Use of flexible endoscopic evaluation of swallow (FEES) is ideal to assist in swallow assessment through direct visualisation of laryngopharyngeal anatomy and function (Hafner et al., 2008; Hales et al., 2008).

It is preferable to assess swallowing with a deflated tracheostomy cuff, as this restores translaryngeal airflow and facilitates sensory awareness. While not optimal, there are no contraindications to oral intake with the tracheostomy cuff inflated. However, due to the higher prevalence of silent aspiration with the cuff inflated (Goff & Patterson, 2018) instrumental assessment is recommended.

15.0 Removal of Tracheostomy (Decannulation)

15.1 Medical Requirements

Medical staff must order decannulation and document plan in the medical record.

15.2 Safety Information

Tracheostomy dilators and appropriate size tracheostomy tubes (same size and one size smaller) must be left at the bedside for 48 hours post de-cannulation.³

Decannulation should not occur after midday[†] or over the weekend unless adequate staff and resources are available to provide assessment and perform advanced airway and tracheostomy management should the decannulation fail.³

Criteria for decannulation:

- Stable arterial blood gases
- Adequate spontaneous breathing effort³ (note increased dead space and increased work of breathing following removal of tube)
- Cuff deflated continuously for a minimum of 24 hours³
- Absence of distress
- Haemodynamic stability³
- Absence of fever or active infection³
- PaCO₂ <60mmHg
- Absence of delirium, adequate mental state^{3,39}
- Adequate swallow^{3,39} and/or normal endoscopic examination indicating lack of occlusion.
- Able to expectorate^{3,26}
- Effective Cough^{3,11,39,42} and gag present
- Ensure no further planned procedure requiring anaesthesia in next 7-10 days^{1,3}

15.3 Nursing Management

- Respiratory status assessment is necessary pre and post removal of the tube.^{39,43}
- Observe patient is breathing adequately through mouth and nose. Watch for good chest expansion and good colour.
- Following removal, the patient should have continuous pulse oximetry for 2 hours with ½ hourly visualisation, followed by every four hours and PRN for a minimum of 24 hours.⁴³

The stoma should heal by secondary intention within 7-10 days for most patients.¹ Failure to close requires medical review, as the patient may require temporary insertion of a mini tracheostomy, or surgical closure^{3,39}.

15.4 Procedural Information

Requirements

- As per mandatory tracheostomy equipment (refer Section 1, Principles of tracheostomy management.)
- PPE⁴³
- Pulse oximetry⁴³
- Dressing pack⁴³
- Occlusive dressing (e.g., Duoderm Thin) Sterile sodium chloride 0.9% solution
- Sterile gauze
- Firm retentive tape (Fixomul®)
- Skin barrier film wipes (i.e., Conveen®) (optional)

Note: additional emergency equipment that must be present during a planned tracheostomy change or decannulation includes⁴³:

- Oxygen and selection of masks including non-rebreather, Hudson and tracheostomy masks, resuscitation mask and bag.
- Intubation and resuscitation equipment (including Bougie or oxygenating intubation catheter) [‡] and a range of tracheal tubes.
- Laryngeal mask
- Spare tracheostomy inner cannula appropriate for tracheostomy.
- Tracheostomy cuff manometer

Procedure

The patient should be fasted for 4 hours or aspirate the nasogastric tube prior to procedure.³

1. Explain the procedure to the patient.
2. Sit patient upright^{3,43}, or semi recumbent.
3. Perform hand hygiene.
4. Gather and assemble equipment as per tracheostomy dressing.
5. Perform hand hygiene, apply PPE.
6. Perform oral, pharyngeal, and tracheal suction.^{3,43}
7. Remove gloves and perform hand hygiene.
8. Apply PPE. Use a 10ml syringe to fully deflate the cuff. (Do not use the manometer to deflate the cuff as this method does not always remove all of the air from the cuff). Remove Velcro tapes.
9. Remove tracheostomy tube by guiding out of stoma gently.
10. Clean stoma using a non-touch technique if secretions are present. Dry well.
11. A skin protective wipe maybe applied to peri-stomal skin prior to dressing application.
12. Place dressing onto the stoma site.
13. Seal dressing securely with retentive tape. (Snip the horizontal sides of tape to give a better fit to neck contours).
14. Monitor dressing seals and ensure no air leak.
15. Observe patient for signs of distress including:
 - Dyspnoea
 - Stridor, laboured/noisy breaths

- Increased respiratory rate and heart rate
 - Excess use of respiratory muscles or diaphragmatic respiration
 - Agitation
 - Oxygen desaturation
16. Discard waste: remove PPE and perform hand hygiene.
 17. Document de-cannulation in inpatient notes immediately following procedure.³
 18. Perform dressing initially daily and PRN.^{3,11}
 19. When secretions are minimal, perform dressings PRN until stoma is sealed.

16.0 Intra / Inter-Hospital Transport of Tracheostomy Patients

16.1 Medical Requirements

Inter hospital transports require medical authorisation and handover to receiving medical team.

16.2 Safety Information

Emergency equipment (listed below) must always accompany the patient.

A nurse escort is required for all tracheostomy patients being transported intra or inter hospital unless exempted by ward CNS.

16.3 Nursing Management

Only a nurse experienced in tracheostomy tube management may escort a patient with a tracheostomy when not accompanied by a medical officer †

For inter-hospital transfers, handover to receiving area must include:

- Complete ISOBAR handover¹¹
- information regarding the need for and type of tracheostomy¹¹ (fenestrated/non-fenestrated, difficulties with insertion, and presence of inner cannula)
- the date of insertion
- secretions (colour, consistency, amount)
- frequency of suctioning, and
- Any relevant instructions for care and equipment required.
- Completed Nursing Transfer Summary (MR 640)

Unplanned and night-time critical care transfer should be avoided or given careful consideration especially in the event of a newly formed stoma or recently weaned from ventilator support. ^[43]

Essential Equipment: The following equipment must be always available during procedures / transfer for a patient with a tracheostomy:

- Portable Oxygen¹¹
- Portable Suction, with appropriately sized catheter attached and Yankauer sucker available¹¹
- Appropriate resuscitation equipment, Resuscitation bag and appropriately sized mask¹¹ (for **inter-hospital** transfer)
- Capnography available in event of emergency¹⁻⁴:
 - Waveform capnography for use; in ICU/HDU
- Catheter mount (Bodai connector)
- Artery forceps
- Tracheostomy wire†
- PPE [non-powdered latex free gloves, apron, face shield]¹¹
- Spare tracheostomy tubes one of the same size and one smaller¹¹

- Tracheal dilators
- Tracheostomy mask¹¹
- Spare Tracheostomy dressing and Velcro tapes
- Cuff manometer and 10mL syringe¹¹
- Stitch cutter and scissors
- Lubricating jelly
- Spare inner cannula in a clean kidney dish.^{2,3}
- Passive humidifier (e.g., HMEF¹¹)

Related Documents

Legislation:

- [Health Practitioner Regulation National Law \(WA\) Act 2010](#)
- [Nursing and Midwifery Board of Australia \(NMBA\) Code of Professional Conduct for Nurses](#)
- [Western Australia: Medicines and Poisons Act 2014,](#)
- [Medicine and Poisons Regulations 2016,](#)
- [Pharmacy Act 2010](#)
- [Work Health and Safety 2020](#)

Policy:

- [NMHS Clinical Documentation policy](#)
- [SCGOPHCG Infection Prevention and Control manual - Aseptic Technique policy.](#)
- [SCGOPHCG Clinical Handover – Communicating for Safety policy](#)
- [SCGOPHCG Patient Identification policy](#)
- [SCGOPHCG Procedure Matching Policy](#)
- Fiona Stanley Fremantle Hospitals Group Policy and Procedure: Tracheostomy Management (Adult). Ref no FSH-HW-POL-0043

Clinical Practice Guidelines:

- [SCGOPHCG Oxygen Therapy CPG](#)
- [SCGOPHCG Nebulisation CPG](#)

Learning resources:

- [Tracheostomy Safety e-Learning:](#)

Departmental Guidelines:

- SCGOPHCG Intensive Care Unit Departmental Guideline: Oral Feeding of Tracheostomised Patients

References

1. Austin Health. Portex Uniperc® Adjustable Flange Tracheostomy Tube Clinical Instruction Sheet. Austin Health; 2022. Accessed November 18, 2025. <https://tracheostomyteam.org/wp-content/uploads/2022/04/Portex-Uniperc-Adjustable-Flange-Tracheostomy-Tube-2.pdf>
2. Austin Health. Use of the Portex Suctionaid® & Blue Line Ultra Tracheostomy Tubes Clinical Instruction Sheet; 2022. Austin Health. Accessed November 18, 2025. <https://tracheostomyteam.org/wp-content/uploads/2022/04/Portex-Suctionaid-BLU-tracheostomy-tube.pdf>
3. Austin Health. The BlueLine Ultra Suctionaid® Tracheostomy Tube: Clinical Procedure; 2023. Austin Health. Accessed November 18, 2025. <https://tracheostomyteam.org/wp-content/uploads/2025/10/USE-OF-THE-SUCTIONAID-TRACHEOSTOMY.pdf>
4. Passy-Muir Inc. Passy-Muir Tracheostomy and Ventilator Swallowing and Speaking Valves Instruction Booklet. Accessed December 16, 2025. [Product Information - Passy-Muir](#)
5. Australian New Zealand Intensive Care Society. Percutaneous Dilatational Tracheostomy Consensus Statement. Australian New Zealand Intensive Care Society; 2014. Accessed November 18, 2025. <https://www.anzics.org/wp-content/uploads/2018/08/2014-The-ANZICS-Percutaneous-Dilatational-Tracheostomy-Consensus-Statement.pdf>

6. Ayhan H, Tastan S, Iyigun E, et al. Normal saline installation before endotracheal suctioning: "What does the evidence say? What do the nurses think?": Multimethod study. *J of Crit Care*. 2015;30(4):762-767. <https://doi.org/10.1016/j.jcrc.2015.02.019>
7. Bevis R, Elenbaas J, Stacey KM. Suctioning: Artificial airway during mechanical ventilation (Respiratory therapy). Elsevier Clinical Skills. 2020; Accessed November 18, 2025. <http://repository.phb.ac.id/677/1/Suctioning-Artificial-Airway-During-Mechanical-Ventilation-Skill-Respiratory-Therapy-030420.pdf>
8. Bodenham A, Bell D, Bonner S, et al. Standards for the Care of Adult Patients with a Temporary Tracheostomy: Standards and Guidelines. Intensive Care Society; 2020. Accessed November 18, 2025. [https://www.ficm.ac.uk/sites/ficm/files/documents/2021-11/2020-08%20Tracheostomy care guidance Final.pdf](https://www.ficm.ac.uk/sites/ficm/files/documents/2021-11/2020-08%20Tracheostomy%20care%20guidance%20Final.pdf)
9. Bonde P, Papachristos I, McCraith A, Kelly B, Wilson C. Sputum retention after lung operation: prospective, randomized trial shows superiority of prophylactic minitracheostomy in high-risk patients. *The Annals of Thoracic Surgery*. 2002;74(1):196-203. [https://doi.org/10.1016/S0003-4975\(02\)03612-3](https://doi.org/10.1016/S0003-4975(02)03612-3)
10. Gillies D, Todd DA, Foster JP, Batuwitage BT. Heat and moisture exchangers versus heated humidifiers for mechanically ventilated adults and children. *Cochrane Database Syst Rev*. 2017 Sep 14;9(9):CD004711. <https://doi.org/10.1002/14651858.CD004711.pub3>
11. Green A, Liddy MA. Specimen collection: Sputum by suction (Respiratory therapy). Elsevier Clinical Skills. 2020; Accessed November 18, 2025. <http://repository.phb.ac.id/676/1/Specimen-Collection-Sputum-by-Suction-Skill-Respiratory-COVID-19-toolkit-21.03.20.pdf>
12. Goff D & Patterson J. Eating and drinking with an inflated tracheostomy cuff: A systematic review of the aspiration risk. *International Journal of Language and Communication Disorders*. 2019;54(1):30–40. <https://doi.org/10.1111/1460-6984.12430>
13. Hafner G, Neuhuber A, Hirtenfelder S, Schmedler B, Eckel HE. Fiberoptic endoscopic evaluation of swallowing in intensive care unit patients. *European Archives of Oto-Rhino-Laryngology*. 2008;265:441–46. <https://doi.org/10.1007/s00405-007-0507-6>
14. Haghighi A, Shafipour V, Bagheri-Nesami M, Baradari A, Charati J. The impact of oral care on health status and prevention of ventilator-associated pneumonia in critically ill patients. *Aus Crit Care*. 2017;30(2):69-73. <https://doi.org/10.1016/j.aucc.2016.07.002>
15. Hales P, Drinnan M, Wilson J. The added value of fiberoptic endoscopic evaluation of swallowing in tracheostomy weaning. *Clinical Otolaryngology*. 2008;33:319–324. <https://doi.org/10.1111/j.1749-4486.2008.01757.x>
16. McGrath BA, Wallace S, Wilson M et al. Effectiveness of above cuff vocalisation for ventilator-dependent patients with tracheostomies. *B J of Anaesthesia*. 2018;120(5): e26. <https://doi.org/10.1016/j.bja.2017.11.059>
17. McGrath BA, Wallace S, Wilson M, et al. Safety and feasibility of above cuff vocalisation for ventilator-dependant patients with tracheostomies. *J of the Intensive Care Society*. 2018;20(1):59-65. <https://doi.org/10.1177/1751143718767055>
18. McGrath BA, Bates L, Atkinson D, Moore JA. Multidisciplinary guidelines for the management of tracheostomy and laryngectomy airway emergencies. *Anaesthesia*. 2012;67(9):1025–41. <https://doi.org/10.1111/j.1365-2044.2012.07217.x>
19. Mitchell RB, Hussey HM, Setzen G, et al. Clinical consensus statement: tracheostomy care. *Otolaryngology Head and Neck Surgery*. Sage Journals. 2012;148(1):6-20. <https://doi.org/10.1177/0194599812460376>
20. Mussa CC, Gomaa D, Rowley DD, et al. AARC Clinical Practice Guideline: Management of adult patients with tracheostomy in the acute care setting. *Respiratory Care*. 2021;66(1):156-169. <https://doi.org/10.4187/respcare.08206>
21. National Tracheostomy Safety Project. NTSP Resources. National Tracheostomy Safety Project; 2025. Accessed November 18, 2025. <https://tracheostomy.org.uk/resources/documents>
22. National Tracheostomy Safety Project. NTSP Algorithms and Bedheads; 2025. Accessed November 18, 2025. <http://www.tracheostomy.org.uk/NTSP-Algorithms-and-Bedheads>
23. NSW Agency for Clinical Innovation. Care of Adult Patients in Acute Care Facilities with a Tracheostomy: Clinical Practice Guideline. NSW Agency for Clinical Innovation; 2021. Accessed November 18, 2025. https://aci.health.nsw.gov.au/_data/assets/pdf_file/0004/685300/ACI-CPG-Care-adult-patients-acute-tracheostomy.pdf
24. Pryor L, Ward E, Cornwell P, O'Connor S & Chapman M. Patterns of return to oral intake and decannulation post tracheostomy across clinical populations in an acute inpatient setting. *International Journal of Language & Communication Disorders*. 2016;51(5):556–567. <https://doi.org/10.1111/1460-6984.12231>
25. Varshney S, Sharma S, Jelly P. Tracheostomy suctioning: Exploration of knowledge and practices of nurses working in selected tertiary care hospitals. *Sage Journals*. 2017;13(2):68-77. <https://doi.org/10.1177/0974150X20170203>
26. Wang C, Tsai J, Chen S, et al. Normal saline instillation before suctioning: A meta-analysis of randomized controlled trials. *Aus Crit Care*. 2017;30(5):260-265. <https://doi.org/10.1016/j.aucc.2016.11.001>

27. Zhao T, Wu X, Zhang Q, Li C, Worthington HV, Hua F. Oral hygiene care for critically ill patients to prevent ventilator-associated pneumonia. Cochrane Database Syst Rev. 2020;12(12):CD008367. doi: <https://doi.org/10.1002/14651858.CD008367.pub4>.

Authorisation & Version Control

Policy Sponsor:	Nurse Co-Director Surgical Services				
Policy Contact:	Clinical Nurse Specialist Intensive Care Unit				
Version #:	8.0	Last Reviewed:	10/02/2026	Review Due:	10/02/2029
Reference #	CPG016				
Approved by:	Nurse Co-Director Acute Division Services SCGOPHCG			Date:	27/03/2026
Endorsed by:	SCGOPHCG Clinical Practice Guideline Subcommittee			Date:	10/02/2026
Clinical Reviewer	K Laurie CNS ICU L Simpson CNS G64				
Research Reviewer	N Maxwell A/Research Nurse, ICU				
Applicable Standards	NSQHS Std 1: Clinical Governance  Std 2: Partnering with Consumers  Std 3: Preventing and Controlling Healthcare Associated Infection  Std 4: Medication Safety  Std 5: Comprehensive Care  Std 6: Communicating for Safety  Std 8: Recognising and Responding to Acute Deterioration 				
Printed or personally saved electronic copies of this document are considered uncontrolled					

[illegible]